Factor Analysis Introduction

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Factor Analysis

Source

- We will briefly talk about exploratory factor analysis or EFA, and defer the confirmatory factor analysis (CFA) for a later date, if time permits.
- These slides are (mostly) based on chapter 8 of Zelterman and chapter 5 of Everitt.
- Everitt is much more detailed: two chapters for EFA and CFA.

PCA and FA

- FA is viewed as being different from PCA
- **PCA**
- is more exploratory
- may not require as many assumptions
- **FA**
- may require additional assumptions (normality)
- often associated with a search for causal models
- rotation is important
- **Big difference is philosophical**
- FA program will do a PCA but not visa versa

Motivation

In many areas of psychology, and other disciplines in the behavioural sciences, often it is not possible to measure directly the concepts of primary interest.

Two obvious examples are intelligence and social class. In such cases, the researcher is forced to examine the concepts indirectly by collecting information on variables that can be measured or observed directly and can also realistically be assumed to be indicators.

The psychologist who is interested in an individual's "intelligence", for example, may record examination scores in a variety of different subjects in the expectation that these scores are dependent in some way on what is widely regarded as "intelligence" but are also subject to random errors.

And a sociologist, say, concerned with people's "social class" might pose questions about a person's occupation, educational background, home ownership, etc., on the assumption that these do reflect the concept he or she is really interested in.

Latent variables

Both “intelligence” and “social class” are what are generally referred to as **latent variables**

latent variables concepts that cannot be measured directly but can be assumed to relate to a number of measurable or **manifest variables**.

The method of analysis most generally used to help uncover the relationships between the assumed latent variables and the manifest variables is **factor analysis.**

The model on which the method is based is essentially that of **multiple regression,** except now the manifest variables are regressed on the unobservable latent variables (often referred to in this context as **common factors**), so that direct estimation of the corresponding regression coeffcients (**factor loadings**) is not possible.

Types of factor analysis

The first is **exploratory factor analysis**, which is used to investigate the relationship between manifest variables and factors without making any assumptions about which manifest variables are related to which factors.

The second is **confirmatory factor analysis** which is used to test whether a specific factor model postulated a priori provides an adequate fit for the covariances or correlations between the manifest variables.

We will first consider the EFA.

Simple example

Spearman considered a sample of children's examination marks in three subjects, Classics (y_1), French (y_2), and English (y_3), from which he calculated the following correlation matrix for a sample of children:

$$\mathbf{R} = \begin{bmatrix} 1.00 & & \\ 0.83 & 1.00 & \\ 0.78 & 0.67 & 1.00 \end{bmatrix}.$$

If we assume a single factor, then the *single-factor model* is specified as follows:

$$\begin{aligned} y_1 &= \lambda_1 f + u_1, \\ y_2 &= \lambda_2 f + u_2, \\ y_3 &= \lambda_3 f + u_3. \end{aligned}$$

In this example, the underlying latent variable or common factor, f, might possibly be equated with intelligence or general intellectual ability.

Explanation

The terms λ_1, λ_2 and λ_3 which are essentially regression coeffcients are, in this context, known as factor loadings.

The terms u_1, u_2, and u_3 represent random disturbance terms and will have small variances if their associated observed variable is closely related to the underlying latent variable.

The variation in u_i actually consists of two parts, the extent to which an individual's ability at Classics, say, differs from his or her general ability and the extent to which the examination in Classics is only an approximate measure of his or her ability in the subject.

In practise, no attempt is made to disentangle these two parts.

The k-factor analysis model

To begin, we assume that we have a set of observed or manifest variables, $\mathbf{y}^\top = (y_1, y_2, \ldots, y_p)$, assumed to be linked to k unobserved latent variables or common factors $f_1, f_2, \ldots, f_m$, where $m < p$, by a regression model of the form

$$
\begin{aligned}
y_1 &= \lambda_{11} f_1 + \lambda_{12} f_2 + \cdots + \lambda_{1m} f_m + u_1, \\
y_2 &= \lambda_{21} f_1 + \lambda_{22} f_2 + \cdots + \lambda_{2m} f_m + u_2, \\
&\vdots \\
y_p &= \lambda_{p1} f_1 + \lambda_{p2} f_2 + \cdots + \lambda_{pm} f_m + u_p.
\end{aligned}
$$

The λ_js are essentially the regression coefficients of the y-variables on the common factors.

The k-factor analysis model

... but in the context of factor analysis these regression coefficients are known as the factor loadings and show how each observed variable, y_i, depends on the common factors.

The factor loadings are used in the interpretation of the factors; i.e., larger values relate a factor to the corresponding observed variables

... and from these we can often, but not always, infer a meaningful description of each factor.

The k-factor analysis model

The regression equations above may be written more concisely as

$$\underbrace{\mathbf{y} - \mu}_{p\times 1} = \underbrace{\mathbf{\Lambda}}_{p\times m} \underbrace{\mathbf{f}}_{m\times 1} + \underbrace{\mathbf{u}}_{p\times 1}. \tag{F1}$$

where

$$\mathbf{\Lambda} = \begin{pmatrix} \lambda_{11} & \dots & \lambda_{1m} \\ \vdots & \ddots & \vdots \\ \lambda_{p1} & \dots & \lambda_{pm} \end{pmatrix}, \quad \mathbf{f} = \begin{pmatrix} f_1 \\ \vdots \\ f_m \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} u_1 \\ \vdots \\ u_p \end{pmatrix}.$$

We assume that 1. the random disturbance terms $u_1, \dots, u_p$ are uncorrelated with each other, and 2. uncorrelated with the factors $f_1, \dots, f_m$.

(The elements of $\mathbf{u}$ are specific to each y_i and hence are generally better known in this context as *specific variates*.)

Assumptions

Specifically, we need to assume:

$$\mathbb{E}f = \mathbf{0} \quad \text{and} \quad \mathrm{Cov}(f) = \mathbb{E}\{ff'\} = \mathbf{I}, \tag{A1}$$

$$\mathbb{E}\mathbf{u} = \mathbf{0} \quad \text{and} \quad \mathrm{Cov}(\mathbf{u}) = \mathbb{E}(\mathbf{u}\mathbf{u}') = \mathbf{\Psi} = \mathrm{Diag}(\Psi_1, \Psi_2, \ldots, \Psi_p). \tag{A2}$$

In addition, the factors (f) are uncorrelated with the errors (u):

$$\mathrm{Cov}(f, \mathbf{u}) = \mathbf{0}. \tag{A3}$$

Together, these assumptions (A1-A3) are sufficient for us to fit the model.

The k-factor analysis model

The two assumptions imply that, given the values of the common factors f, the manifest variables y are independent; that is, the correlations of the observed variables arise from their relationships with the common factors.

We also assume that factors are uncorrelated with one another, in which case the factor loadings are the correlations of the manifest variables and the factors.

Model

These assumptions can also be used to describe an alternative form for the factor analysis model. If we begin at [(1)], then we can write

$$(y-\mu)(y-\mu)' = (\mathbf{\Lambda} f + \mathbf{u})(\mathbf{\Lambda} f + \mathbf{u})'$$

$$= \mathbf{\Lambda} f(\mathbf{\Lambda} f)' + \mathbf{u}(\mathbf{\Lambda} f)' + \mathbf{\Lambda} f \mathbf{u}' + \mathbf{u}\mathbf{u}'.$$

Using the three assumptions A1-A3, we can then express the covariance matrix ( ) of $\mathbf{y}$ as

$$\mathrm{Cov}(\mathbf{y}) = \mathbb{E}(\mathbf{y}-\mu)(\mathbf{y}-\mu')$$

$$= \mathbf{\Lambda}\mathbb{E}(ff')\mathbf{\Lambda}' + \mathbb{E}(\mathbf{u}f')\mathbf{\Lambda}' + \mathbf{\Lambda}\mathbb{E}(f\mathbf{u}') + \mathbb{E}(\mathbf{u}\mathbf{u}')$$

$$= \mathbf{\Lambda}\mathbf{\Lambda}' + \mathbf{\Psi} \tag{F2}$$

demonstrating the equivalence of two formulations of the factor analysis models at [F1] and [F2].

Communality and unique variance

- Under all these assumptions, the factor analysis model (F2) implies that the variance of variable y_i can be written as:

$$\sigma_i^2 = \underbrace{\sum_{j=1}^{p} \lambda_{ij}^2}_{\text{communality}} + \overbrace{\psi_i}^{\text{specific variance}}$$

- So, variance of each observed variable can be split into two parts:

1. **Communality**, represents the variance shared with the other variables via the common factors.
2. **Specific** or **unique variance** and relates to the variability in y_i not shared with other variables

Covariance

- The factor model leads to the following expression for the covariance of variables y_i and y_j:

$$\sigma_{ij} = \sum_{l=1}^{p} \lambda_{il}\lambda_{jl}$$

- The covariances are not dependent on the specific variates in any way; it is the common factors only that aim to account for the relationships between the manifest variables.

Rescaling the variables

- Suppose, we re-scale the variables to $\mathbf{z} = \mathbf{C}\mathbf{y}$, where $\mathbf{C} = diag(c_i)$.
- We can show that the factor analysis model holds for $\mathbf{z}$ with factor loading matrix $\Lambda_z = C\Lambda_y$ and $\Psi_z = C\Psi C = diag(c_i^2\psi_i)$.
- The factor loading matrix for the scaled variables y is obtained by multiplying the ith row of $\mathbf{y}$ by c_i, and the same applies to specific variances.
- If the scaling factors are $c_i = 1/s_i$ (where s_i is the standard deviation of y_i), the factor analysis model effectively applies to the correlation matrix instead of the covariance matrix.
- As a result, factor analysis can be conducted using either the covariance or correlation matrix, yielding equivalent outcomes.

R functions

- In `R`, we use `factanal(x,factors=m)` where `x` is the data arranged in the usual fashion as columns of variables and rows for observations.
- The `m` parameter specifies how many factors we want `R` to fit.
- Usually, we try to fit a relatively small number of factors (at most three or four) because of the difficulty of their interpretation.
- There is also a formal statistical hypothesis test performed in factanal() testing whether m factors are adequate or not.

Solutions?

Two main methods:

1. principal factor analysis, and
2. maximum likelihood factor analysis,

Principal Factor method

- Try to ignore the diagonal elements. Replace with an estimate of the common variance. Usually the squared multiple correlation between one variable and the others is used.
- The method may also be iterated.

Maximum Likelihood method

- fits the model by fitting the covariance/correlation structure using the likelihood function for the multivariate normal distribution.
- The likelihood involves the variance covariance matrix which may be written as a function of the loadings and specific variances.
- Requires iterative solution

Hemangioma example

- Most common childhood cancer diagnosis. It appears as a lump of blood vessels on the skin. If left untreated it usually resolves within a few years. Some parents opt for surgical removal.
- Data: measurements on several genetic markers thought to be related to the disease.
- The age, measured in days, is the infant's age at the time of the surgery, i.e., how long the infant was exposed to the tumor.
- It was hypothesized exposure to the tumor might influence the magnitude of the expression of some of these genes over time.

Hemangioma Data

Table 8.2 Age, in days. and expression of genetic markers for infants who were surgically treated for hemangioma

Age	RB	p16	DLK	Nanog	C-Myc	EZH2	IGF-2
81	2.0	3.07	308975	94	6.49	2.76	11176
95	6.5	1.90	70988	382	1.00	7.09	5340
95	3.6	3.82	153061	237	0.00	5.57	6310
165	1.9	3.74	596992	88	0.00	2.47	7009
286	2.6	5.17	369601	282	12.23	1.63	7104
299	2.9	5.76	1119258	177	8.76	3.51	9342
380	1.9	2.40	214071	45	5.76	1.41	3726
418	7.1	3.38	69511	265	1.17	3.07	8039
420	6.4	3.37	81457	659	1.88	3.87	12583
547	6.4	4.05	64348	336	0.78	4.76	6505
590	1.8	5.15	164881	2012	35.65	9.45	32722
635	6.7	2.67	126016	3072	0.00	4.35	11763
752	1.8	3.28	567858	127	4.13	1.00	10283
760	7.3	0.92	43438	698	1.77	3.32	11518
1171	1.8	6.56	716260	392	12.92	2.90	13264
1277	1.3	0.05	94	15	0.36	3.83	30
1520	4.0	2.79	31125	454	0.62	2.33	1163
2138	0.5	0.00	2331	33	0.03	0.17	66
3626	4.2	4.24	560208	340	5.43	1.36	21174

Data courtesy of Dr. Deepak Nayaran

Hemangioma example

Hemangioma example

```r
hemangioma <- read.csv("hemangioma_data.csv")
fa.out <- factanal(hemangioma, factors = 3)
fa.out$loadings
```

```
Loadings:
      Factor1 Factor2 Factor3
Age                   -0.168
RB            -0.170   0.964
p16    0.381   0.747
DLK            0.971  -0.211
Nanog  0.475           0.389
C.Myc  0.930   0.280  -0.227
EZH2   0.614  -0.115   0.343
IGF.2  0.767   0.351   0.187

               Factor1 Factor2 Factor3
SS loadings      2.225   1.751   1.364
Proportion Var   0.278   0.219   0.170
Cumulative Var   0.278   0.497   0.668
```

We examine the three-factor loadings to see `(C.Myc, EZH2, IGF-2)` collectively appear to work together in the first factor. The second factor is almost entirely the `DLK` marker. The third factor is almost represented by the `RB` marker. `Age` plays almost no part, contrary to our original hypothesis.

Uniqueness

- Recall the factor model:

$$\Sigma = \mathbf{\Lambda}\mathbf{\Lambda}' + \mathbf{\Psi} \tag{F2}$$

- The uniqueness, sometimes referred to as noise, corresponds to the proportion of variability, which can not be explained by a linear combination of the factors. This is the Ψ in (F2).
- A high uniqueness for a variable indicates that the factors do not account well for its variance.

```
fa.out$uniquenesses
```

```
       Age         RB        p16        DLK      Nanog      C.Myc       EZH2 
0.96312288 0.03388095 0.29067961 0.00500000 0.61640634 0.00500000 0.49146204 
     IGF.2 
0.25418248 
```

Loadings, again

- Loadings range from -1 to 1, this is the Λ.
- The loadings are the contribution of each original variable to the factor. Variables with a high loading are well explained by the factor.
- Notice there is no entry for certain variables. That is because R does not print loadings less than 0.1

```
Loadings:
      Factor1 Factor2 Factor3
Age                   -0.168
RB            -0.170   0.964
p16    0.381   0.747
DLK            0.971  -0.211
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```

Communality and Specific variance

- Recall the decomposition:

$$\sigma_i^2 = \underbrace{\sum_{j=1}^{p} \lambda_{ij}^2}_{\text{communality}} + \overbrace{\psi_i}^{\text{specific variance}} \quad \Rightarrow \quad \sum_{j=1}^{p} \lambda_{ij}^2/\sigma_i^2 + \psi_i/\sigma_i^2 = 1.$$

Communality and Specific variance

- Here, the $\sum_j \lambda_{ij}^2/\sigma_i^2$ gives the fraction of the variable's total variance explained by the factor $\sim$ communality.
- We calculate the communality by subtracting the uniquenesses from 1

```r
apply(fa.out$loadings^2, 1, sum) # communality
```

```
       Age         RB        p16        DLK      Nanog      C.Myc       EZH2 
0.03688578 0.96611902 0.70931820 0.99500266 0.38359679 0.99500472 0.50853609 
     IGF.2 
0.74581358 
```

```r
1 - apply(fa.out$loadings^2, 1, sum) # uniqueness
```

```
        Age          RB         p16         DLK       Nanog       C.Myc 
0.963114225 0.033880978 0.290681797 0.004997344 0.616403209 0.004995279 
       EZH2       IGF.2 
0.491463908 0.254186419 
```

- An appropriate factor model results in low values for uniqueness and high values for communality.

Proportion of variance explained

- The table below the loadings shows the proportion of variance explained by each factor, with cumulative variance (`Cumulative Var`) ranging from 0 to 1.
- The row `Proportion Var` indicates the variance explained by each factor, while `SS loadings` gives the sum of squared loadings.
- A factor is considered valuable if its `SS loadings` exceeds 1 (Kaiser's rule!)

```
Loadings:
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Age                   -0.168
RB            -0.170   0.964
p16    0.381   0.747
DLK            0.971  -0.211
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```

Testing

- The last section of the function output shows the results of a hypothesis test.
- H_0: the number of factors in the model, in our case $ m = 3$ factors, is sufficient.
- Usually, we reject H_0 if the p-value is less than 0.05, and say that the number of factors is too small.

```
fa.out$PVAL
```

```
objective
0.3300475
```

P-values

- We can calculate the P-values for a sequence of number of factors, as shown below:

```r
sapply(1:4, function(f) factanal(hemangioma, factors = f, method ="mle")$PVAL)
```

```
  objective   objective   objective   objective
0.007295322 0.062215517 0.330047536 0.519102928
```

Full output

```r
fa.out
```

```
Call:
factanal(x = hemangioma, factors = 3)

Uniquenesses:
  Age    RB   p16   DLK Nanog C.Myc  EZH2 IGF.2 
0.963 0.034 0.291 0.005 0.616 0.005 0.491 0.254 

Loadings:
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```
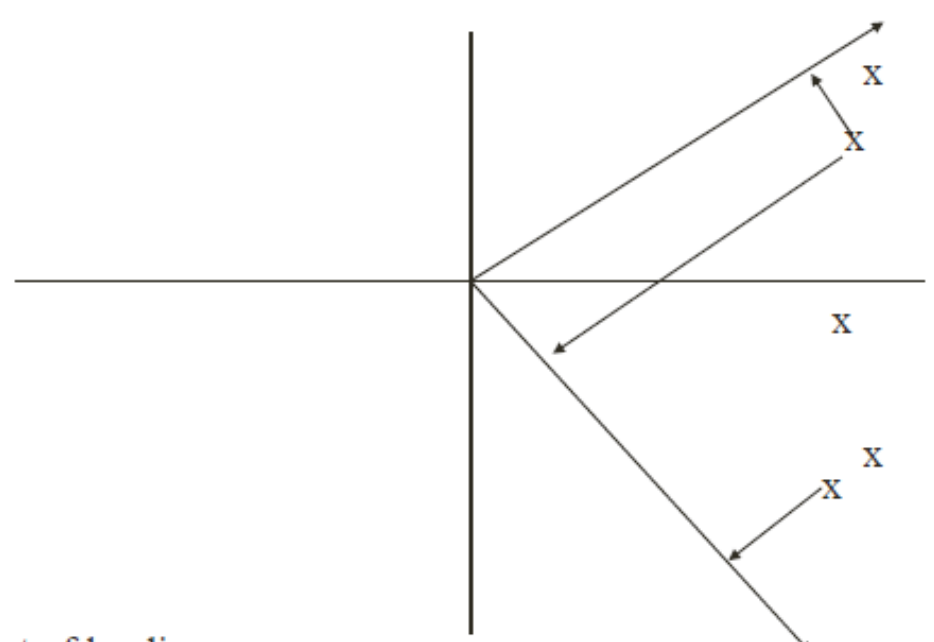

Rotations!

- Up until now, we have conveniently ignored one problematic feature of the factor analysis model, namely that, there is no unique solution for the factor loading matrix.
- That is, the loadings are only **unique up to a rotation**
- That is, you can multiply the factors f by an orthogonal matrix $M_{p \times p}$ without changing the variance decomposition.



Plot of loadings

Simple structure and rotation

- The solution from PCA and other extraction methods focus on “fit to the correlation” matrix not on explanation.
- It was observed early that one could rotate the solution (that is rotate the loadings matrix) and not change the fit.
- Hence we look at rotation as a way to make the solution **better**.
- This leads to simple structure as a definition of better.

1. Most loadings on any factor should be small and only a few should be large in absolute value. A factor only has a few high loading.
2. A specific row of the loadings matrix Λ should display nonzero loadings on only one or at most a few factors. A variable loads on only a few factors
3. Any pair of factors should have different patterns of loadings. Unique factors.

To be continued

Factor Analys

Recap: Model

The factor analysis model has two equivalent representation:

$$\underbrace{\mathbf{y} - \mu}_{p\times 1} = \underbrace{\mathbf{\Lambda}}_{p\times m} \underbrace{\mathbf{f}}_{m\times 1} + \underbrace{\mathbf{u}}_{p\times 1}. \tag{F1}$$

where

$$\mathbf{\Lambda} = \begin{pmatrix} \lambda_{11} & \dots & \lambda_{1m} \\ \vdots & \ddots & \vdots \\ \lambda_{p1} & \dots & \lambda_{pm} \end{pmatrix}, \quad \mathbf{f} = \begin{pmatrix} f_1 \\ \vdots \\ f_m \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} u_1 \\ \vdots \\ u_p \end{pmatrix}.$$

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Assumptions

Specifically, we need to assume:

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In addition, the factors (f) are uncorrelated with the errors (u):

$$\mathrm{Cov}(f, \mathbf{u}) = \mathbf{0}. \tag{A3}$$

Together, these assumptions (A1-A3) are sufficient for us to fit the model.

Alternative formulation

- Under (A1-A3), we can show that:

$$(y-\mu)(y-\mu)' = (\mathbf{\Lambda} f + \mathbf{u})(\mathbf{\Lambda} f + \mathbf{u})'$$

$$= \mathbf{\Lambda} f(\mathbf{\Lambda} f)' + \mathbf{u}(\mathbf{\Lambda} f)' + \mathbf{\Lambda} f \mathbf{u}' + \mathbf{u}\mathbf{u}'.$$

Using the three assumptions A1-A3, we can then express the covariance matrix ( ) of $\mathbf{y}$ as

$$\mathrm{Cov}(\mathbf{y}) = \mathbb{E}(\mathbf{y}-\mu)(\mathbf{y}-\mu')$$

$$= \mathbf{\Lambda}\mathbb{E}(ff')\mathbf{\Lambda}' + \mathbb{E}(\mathbf{u}f')\mathbf{\Lambda}' + \mathbf{\Lambda}\mathbb{E}(f\mathbf{u}') + \mathbb{E}(\mathbf{u}\mathbf{u}')$$

$$= \mathbf{\Lambda}\mathbf{\Lambda}' + \mathbf{\Psi} \tag{F2}$$

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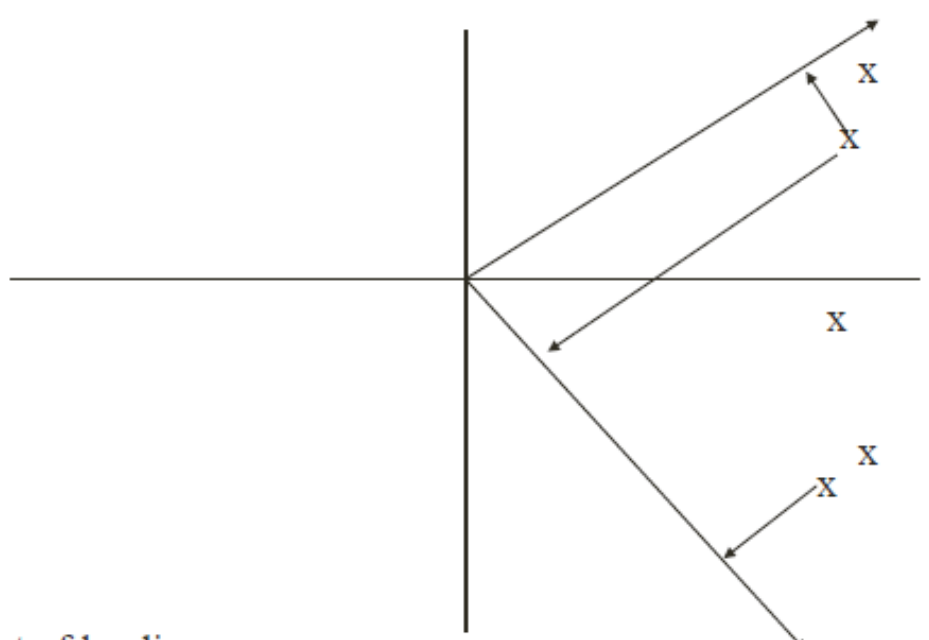

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Rotation

- Most users of factor analysis rotate the solution to try to get factors which display simple structure.
- This makes them easier to interpret.
- Rotation (orthogonal) does not change the distance properties of the loadings.
- Hence total variance and variance explained for each variable remains unchanged.
- However it is now distributed differently amongst the factors.

Why invariant?

- Suppose we have an orthogonal matrix: M such that $MM^\top = M^\top M = I$.
- Then, we can modify a factor model by introducing the factor M that won't change the covariance.

$$\mathbf{y} = (\mathbf{\Lambda M})(\mathbf{M}^\top \mathbf{f}) + \mathbf{u}$$

- New factors: $M^\top f$ and new loadings ΛM.
- The covariance matrix of the observed:

$$\begin{aligned} \mathbf{\Sigma} &= (\mathbf{\Lambda M})(\mathbf{\Lambda M})^\top + \mathbf{\Psi} \\ &= \mathbf{\Lambda}(\mathbf{M}^\top \mathbf{M})\mathbf{\Lambda}^\top + \mathbf{\Psi} \\ &= \mathbf{\Lambda\Lambda}^\top + \mathbf{\Psi} \end{aligned} \tag{F2}$$

- Same covariance structure!

Factor rotation

- It may be that a more interpretable orthogonal solution can be achieved using the equivalent model with loadings $\mathbf{\Lambda}^* = \mathbf{\Lambda}\mathbf{M}$ for some particular orthogonal matrix, $\mathbf{M}$.
- Such a process is generally known as factor rotation.

Criticism of Factor Analysis

- Factor analysis has faced criticism due to the flexibility of factor rotation.
- Some argue that rotation allows researchers to manipulate results to fit preconceived notions (Blackith & Reyment, 1971).
- However, these concerns are largely unjustified—rotation is a mathematical tool, not a way to change the underlying structure.

Purpose of Factor Rotation

- Rotation simplifies factor analysis results, making them more interpretable.
- It does **not** alter the fundamental structure of the solution.
- A well-rotated solution:
  - Has each variable loading strongly on only **one** factor.
  - Avoids bipolar factors and intermediate loadings.
  - Groups variables into distinct sets linked to specific factors.

Simple Structure in Factor Analysis

Thurstone (1931) defined **simple structure** as a desirable property of factor solutions, characterized by:

- Each row of the factor loading matrix has at least **one zero**.
- Each column of the loading matrix has at least **m zeros**.
- Every pair of factor columns has variables with **zero loadings in one but not the other**.
- With four or more factors, most variables should have **zero loadings in both columns**.
- Only a **few variables** should have nonzero loadings in multiple factors.

This structure enhances interpretability, making it easier to distinguish between factors.

Example of simple structure

Achieving Simple Structure

- **Goal:** Observed variables should form distinct groups, loading **high on one factor** and **low/negligible on others**.
- Medium-sized, ambiguous loadings should be avoided.
- Simple structure is sought **after** determining:
  1. The number of common factors.
  2. The communalities of each variable.

Rotation and Factor Independence

- Factor loadings are adjusted via **post-multiplication by an orthogonal matrix**.
- This is equivalent to a **rigid rotation** of factor axes.
- Initially, factors are assumed to be **orthogonal (independent)** for simplicity.
- During rotation, we may **relax** this assumption, allowing for **correlated factors** (oblique rotation).

Types of Rotation

- Two types of factor rotation:
  - **Orthogonal rotation**: Keeps rotated factors **uncorrelated**.
  - **Oblique rotation**: Allows factors to be **correlated**.
- **Orthogonal rotation** is achieved by post-multiplying the original loadings matrix by an **orthogonal matrix**.
- **Oblique rotation** uses a matrix that is **not constrained** to be orthogonal.

Choosing a Rotation Method

- The first decision when rotating factors is:
  Should the rotation be orthogonal or oblique?
- There is **no universal answer** to this question.
- Each method has advantages and disadvantages.
- **Oblique rotation** is preferred when the goal is to **best fit the data**.
- **Orthogonal rotation** is preferred when the goal is **generalizability**.

Advantages of Orthogonal Rotation

- **Simpler interpretation**:
  - Loadings represent **correlations** between factors and observed variables.
- **Easier to compare across studies**.
- **No correlations** between factors, making results more generalizable.

Considerations for Oblique Rotation

- **Factors can be correlated**, which is **often more realistic** in real-world data.
- However:
  - Loadings **do not directly** represent simple correlations.
  - Interpretation may be **more complex** due to factor correlations.

Common Rotation Techniques

- **Orthogonal Rotation**
  - Varimax
  - Quartimax
- **Oblique Rotation**
  - Oblimin
  - Promax

Varimax Rotation (Orthogonal)

- Proposed by **Kaiser (1958)**.
- Aims for **factors with a few large loadings** and as many near-zero loadings as possible.
- Achieved by **iterative maximization** of a quadratic function of the loadings (see **Mardia et al. (1979)**).
- Produces factors that:
  - Have **high correlations** with a **small set of variables**.
  - Have **little to no correlation** with other variables.
- **Tends to eliminate** general factors by redistributing factor variance.

Quartimax Rotation (Orthogonal)

- Introduced by **Carroll (1953)**.
- Encourages **each variable** to correlate highly with **one factor** and weakly (or not at all) with others.
- Less commonly used than **varimax**.

Oblimin Rotation (Oblique)

- Developed by **Jennrich & Sampson (1966)**.
- Seeks **simple structure** in the **factor pattern matrix**.
- Includes a **tuning parameter** to **control the degree of factor correlation**.
- **Choosing the parameter value is difficult**, but **Pett, Lackey, & Sullivan (2003)** suggest values between **−0.5 and 0.5** for most applications.

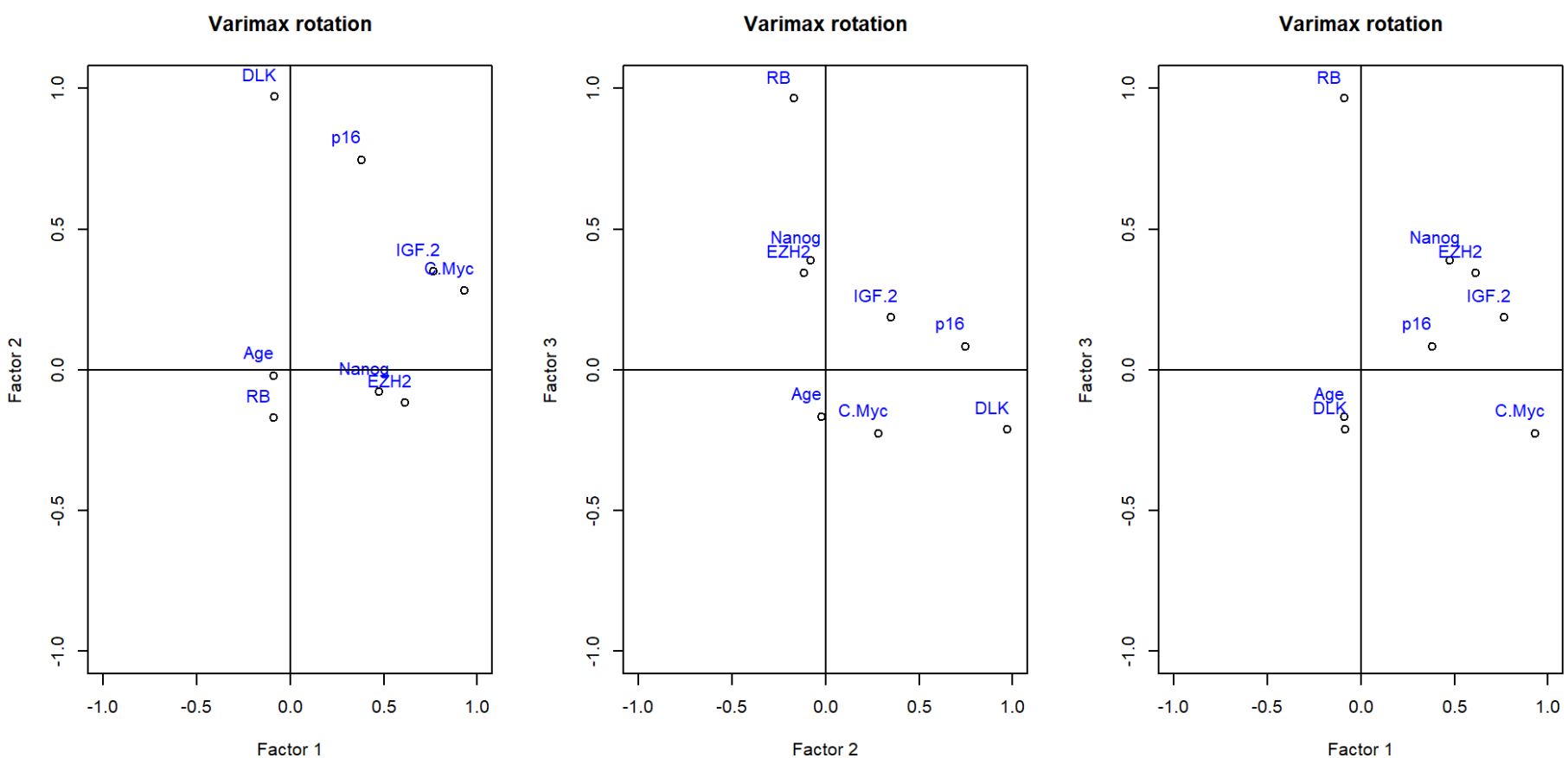
Promax Rotation (Oblique)

- Introduced by **Hendrickson & White (1964)**.
- Works by **raising the loadings** from an initial **orthogonal solution** (often varimax) to a power.
- Goal:
  - **Best structure** with **lowest power loadings**.
  - **Minimal correlation** between factors.

Varimax

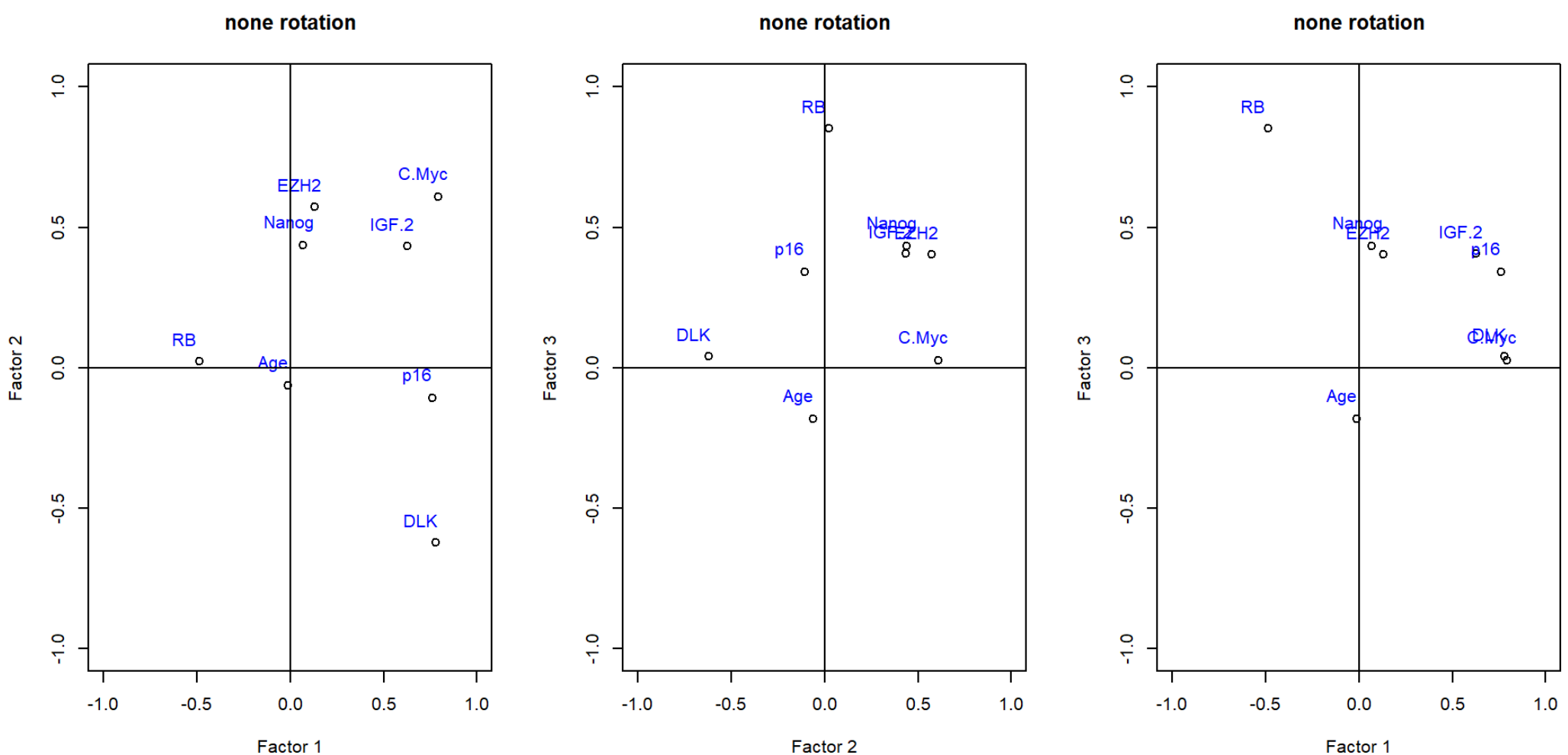
- The default rotation is “varimax” in `factanal()` function which is one of the orthogonal rotation methods.

```
heme_varimax <- factanal(hemangioma, factors = 3, rotation = "varimax")
```



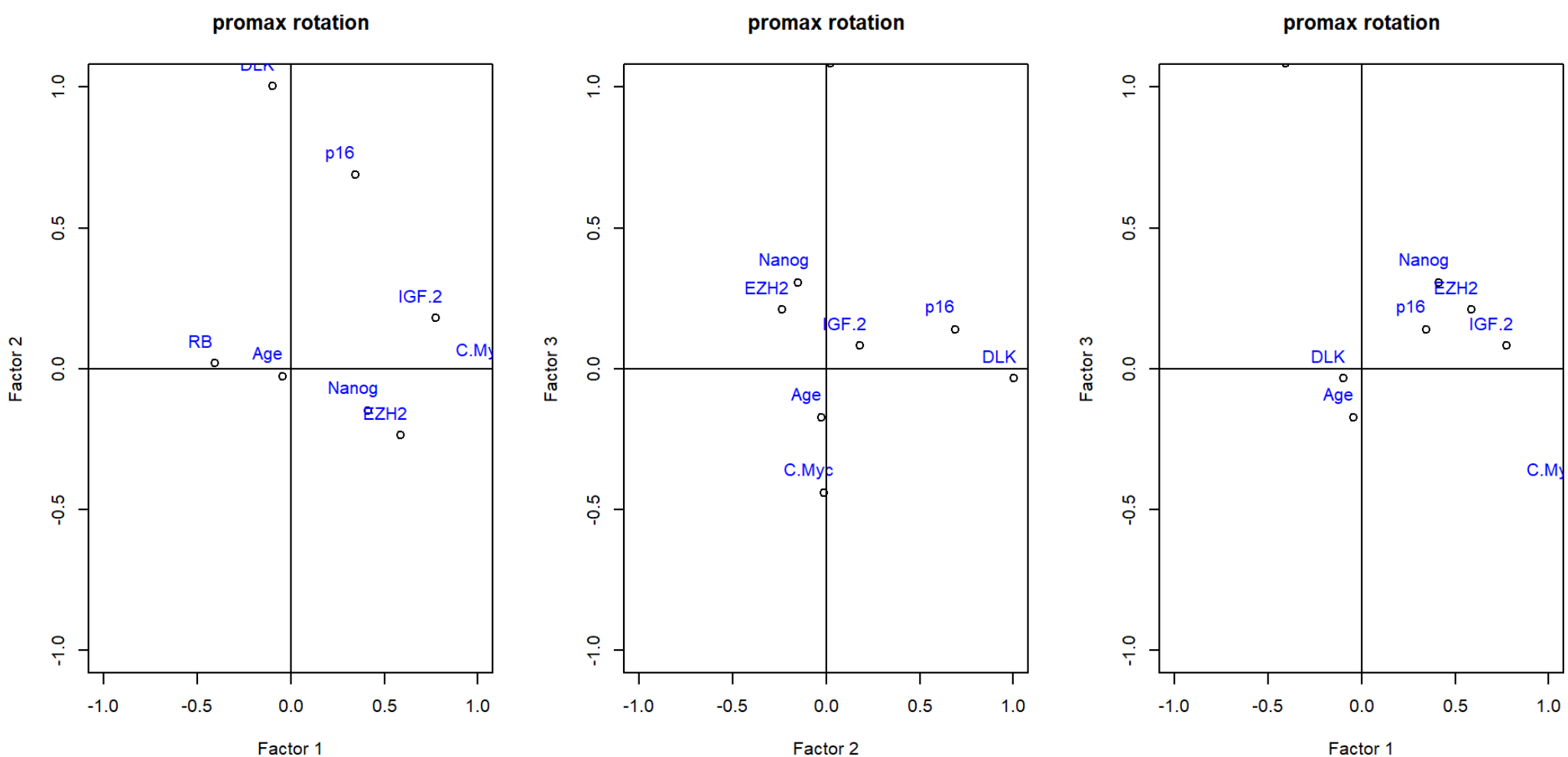
No rotation

```
heme_none <- factanal(hemangioma, factors = 3, rotation = "none")
```



Promax rotation

```
heme_promax <- factanal(hemangioma, factors = 3, rotation = "promax")
```



Other R packages

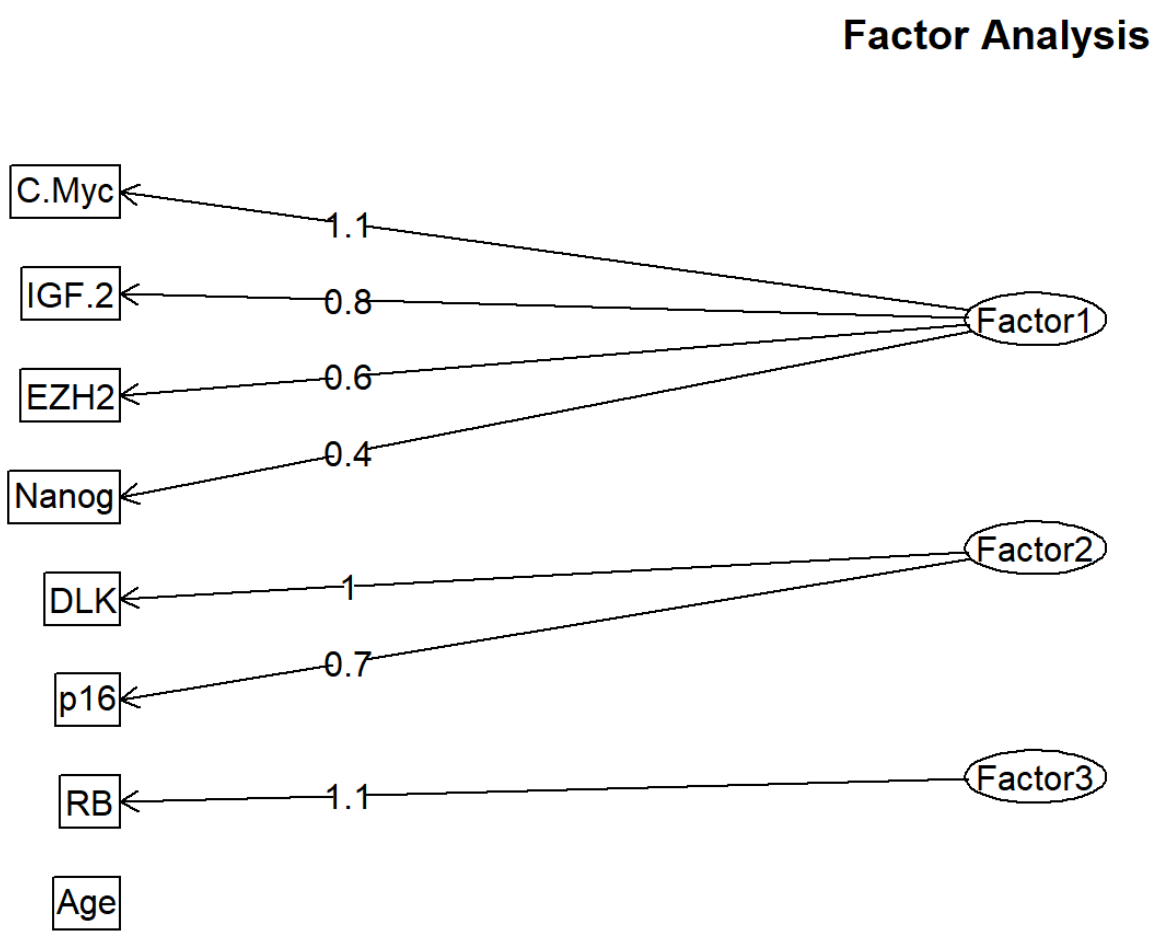
- psych is a very nice package with lots of tools and functions. The following code does factor analysis for the hemangioma data with varimax rotation.

```r
library(psych)
heme_psych_fa <- fa(hemangioma, # input data
               nfactors = 3, # number of factors
               rotate = "varimax") # rotation

heme_psych_fa$loadings # factor loadings
```

```
Loadings:
      MR1    MR2    MR3
Age                -0.735
RB     0.103 -0.441  0.248
p16    0.405  0.700  0.229
DLK           0.865
Nanog  0.611 -0.282  0.134
C.Myc  0.721  0.381
EZH2   0.631 -0.242  0.551
IGF.2  0.967  0.215 -0.125

                 MR1   MR2   MR3
SS loadings    2.402 1.762 0.995
Proportion Var 0.300 0.220 0.124
Cumulative Var 0.300 0.521 0.645
```

Diagrams

```
library(psych)
heme_loads <- heme_promax$loadings
fa.diagram(heme_loads)
```



The `psych` package

- Supports more functionality. For example, the `rotate` argument has the following description if you look at the help for `fa`.

> "none", "varimax", "quartimax", "bentlerT", "equamax", "varimin", "geominT" and "bifactor" are orthogonal rotations. "Promax", "promax", "oblimin", "simplimax", "bentlerQ", "geominQ" and "biquartimin" and "cluster" are possible oblique transformations of the solution. The default is to do a oblimin transformation, although versions prior to 2009 defaulted to varimax.

Other

- GPArotation

> GPArotation package contains functions for the rotation of factor loadings matrices. The functions implement Gradient Projection (GP) algorithms for orthogonal and oblique rotation. Additionally, a number of rotation criteria are provided. The GP algorithms minimize the rotation criterion function, and provide the corresponding rotation matrix. For oblique rotation, the covariance / correlation matrix of the factors is also provided.

SOME ASSESSMENTS

- **Error matrix** - Compute the estimated variance covariance matrix then compare it with the original matrix or compute the residual matrix. Like to have the off diagonals small
- **Partial correlation matrix** - correlation matrix between the variables after partialling out the information provided by the factors. For example, the correlation between y1 and y2 adjusted for factor 1. These values should be near zero. (There is a test Kaiser-Meyer-Olkin test for adequacy using the partial correlations).

KMO test

Henry Kaiser (1970) introduced an Measure of Sampling Adequacy (MSA) of factor analytic data matrices.

"In his delightfully flamboyant style, Kaiser (1975) suggested that KMO > .9 were marvelous, in the .80s, mertitourious, in the .70s, middling, in the .60s, medicore, in the 50s, miserable, and less than .5, unacceptable."

```
KMO(hemangioma)
```

```
Kaiser-Meyer-Olkin factor adequacy
Call: KMO(r = hemangioma)
Overall MSA =  0.51
MSA for each item =
  Age    RB   p16   DLK Nanog C.Myc  EZH2 IGF.2
 0.30  0.27  0.56  0.43  0.84  0.48  0.72  0.55
```

Terminology note

- **Loadings** – associated with correlations with variables and weights, usually applied in orthogonal factor model.
- **Pattern** – estimates of weights in factor analysis – analogous to the coefficients in regression (Λ in the model) (preferred over structure coeffs). Replaces the loadings matrix in oblique rotation. Squares represent unique contribution of factor to variable.
- **Factor Structure** – correlation between the factors and variables, more commonly applied in oblique situations (same as pattern in orthogonal but not oblique due to correlation between factors). Represent unique contribution from factor and contribution from correlation with other factors.
- Pattern usually reported but may not be good when there are highly correlated factors. Analogous to reporting $\beta(x_i)$ in regression versus $corr(y, x_i)$

Comments

Different methods may yield considerably different solutions and rotated solutions.

Rotation may obscure a dominant factor and is influenced by the number of factors retained.

Rotation should follow the maximum likelihood solution (as the constraint was added to insure uniqueness of a solution).

With PCA, rotation destroys the variance maximizing property.

The total variance explained becomes spread over the factors. For example, factor 1 may not have the largest variance.

Factor Analysis vs. PCA

- Both **reduce dimensionality**, but differ in approach:
  - **Factor Analysis** explains **covariances/correlations** using **common factors**.
  - **PCA** explains **variance** in observed variables.
- Adding a new **PCA component** (**m** → **m+1**) leaves prior components unchanged, but adding a factor in **Factor Analysis** can alter all factors.
- **PCA scores** are straightforward to compute; **Factor scores** require more complex methods.

Covariance, Correlation, and Interpretation

- **PCA:** Principal components from **correlation vs. covariance matrices** are generally **different**.
- **Factor Analysis (ML):** Results from **correlation vs. covariance matrices** are **essentially equivalent**.
- **Rotation in Factor Analysis** helps interpretability but does not change the mathematical structure.

COMMENTS - SAMPLE SIZE

- No hard rules
- but with good factors
  - at least 5 observations per variable
  - at least 100 observations
- More observations required for weaker relationships $n > 10 \times p$.

COMMENTS/ STRATEGIES

- Try more than one method to see if results are consistent!
- If the data is large enough, try splitting the data in half to see if the solutions are similar.
- The choice of the number of factors is critical when the solution is rotated. Try dropping or adding a factor to see how stable the rotated solution is.
- Plot the data if possible – look for linearity and normality Remove variables that are linear combinations of other variables.
- These cause problems with ML methods Remove variables that represent single factors (factors that are made of only one variable).
- These represent solitary factors that may confuse rotation methods. You can of course look at these but worry about the effect of keeping them

Envoi

5.11 Summary

Factor analysis has probably attracted more critical comments than any other statistical technique. Hills (1977), for example, has gone so far as to suggest that factor analysis is not worth the time necessary to understand it and carry it out. And Chatfield and Collins (1980) recommend that factor analysis should not be used in most practical situations. The reasons for such an openly sceptical view about factor analysis arise first from the central role of latent variables in the factor analysis model and second from the lack of uniqueness of the factor loadings in the model, which gives rise to the possibility of rotating factors. It certainly is the case that, since the common factors cannot be measured or observed, the existence of these hypothetical variables is open to question. A factor is a construct operationally defined by its factor loadings, and overly enthusiastic reification is not recommended. And it is the case that, given one factor loading matrix, there are an infinite number of factor loading matrices that could equally well (or equally badly) account for the variances and covariances of the manifest variables. Rotation methods are designed to find an easily interpretable solution from among this infinitely large set of alternatives by finding a solution that exhibits the best simple structure.

Factor analysis can be a useful tool for investigating particular features of the structure of multivariate data. Of course, like many models used in data analysis, the one used in factor analysis may be only a very idealised approximation to the truth. Such an approximation may, however, prove a valuable starting point for further investigations, particularly for the confirmatory factor analysis models that are the subject of Chapter 7.